

RUSSELL, KS, USA

WMO: 724585

Lat: 38.876N

Lon: 98.809W

Elev: 568

StdP: 94.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.7	-20.6	0.6	-14.9	-17.9	0.8	-12.0	15.0	3.4	12.9	3.4	4.7	330	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	38.1	22.3	36.4	22.4	34.6	22.3	24.7	32.6	24.0	31.9	23.3	31.3	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.7	27.7	22.0	17.9	27.0	21.2	17.0	26.5	78.0	32.4	75.1	31.9	72.3	31.1	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	12.2	11.1	-20.2	40.8	2.9	1.9	-22.3	42.1	-24.1	43.3	-25.7	44.3	-27.8	45.7
			-20.6	26.1	2.9	1.3	-22.7	27.0	-24.3	27.8	-25.9	28.5	-28.0	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	-0.9	1.0	6.7	12.0	17.8	24.0	26.8	25.4	20.9	13.5	6.2	0.1
	DBStd	11.09	6.06	6.83	6.47	5.46	4.83	4.05	3.42	3.27	4.86	5.33	5.48	5.79
	HDD10.0	1253	338	258	141	40	2	0	0	0	1	26	137	310
	HDD18.3	2845	596	485	362	200	71	5	1	1	29	166	363	567
	CDD10.0	2294	1	7	37	99	244	421	521	478	328	133	23	2
	CDD18.3	846	0	0	1	9	54	176	263	220	106	16	0	0
	CDH23.3	9737	0	3	35	157	613	2026	3206	2377	1111	199	10	0
	CDH26.7	4886	0	1	5	46	237	1023	1797	1214	500	63	1	0
Wind	WSAvg	5.6	5.1	5.5	6.0	6.5	5.7	5.8	5.3	4.9	5.5	5.6	5.5	5.2
Precipitation	PrecAvg	667	14	20	53	65	94	89	88	85	70	42	26	20
	PrecMax	1022	49	70	236	200	275	240	292	190	304	125	116	81
	PrecMin	304	0	0	1	5	3	12	5	17	3	4	0	0
	PrecStd	164	12	18	47	40	58	58	64	44	61	33	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.5	27.0	31.2	35.1	39.3	40.8	38.8	37.2	32.1	24.8	18.1
		MCWB	8.5	10.8	13.8	16.6	19.6	22.0	22.6	22.2	21.3	18.5	13.7	10.2
	2%	DB	14.5	18.4	23.7	27.6	31.8	36.6	38.9	36.9	34.2	28.0	21.3	14.4
		MCWB	6.7	9.7	13.4	15.8	20.0	22.2	22.5	22.5	21.2	17.0	12.0	7.9
	5%	DB	11.2	14.6	20.6	24.5	29.3	34.4	37.0	35.0	32.0	25.2	18.1	11.7
		MCWB	5.3	7.9	11.8	15.0	19.3	22.1	22.6	22.6	20.7	15.8	10.8	6.2
	10%	DB	8.1	11.4	17.4	21.7	26.8	32.4	34.9	33.0	29.5	22.7	15.5	8.7
		MCWB	3.8	6.1	11.2	13.8	18.6	21.7	22.7	22.3	19.8	15.1	9.7	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	12.9	17.3	19.1	22.9	25.1	25.8	25.6	23.8	21.2	15.6	12.7
		MCDB	16.0	18.8	22.5	26.0	29.6	33.4	33.6	32.9	30.9	27.5	20.4	15.3
	2%	WB	7.3	10.6	15.4	17.7	21.8	24.0	24.8	24.5	22.7	19.3	13.9	8.7
		MCDB	13.5	16.4	19.5	24.3	28.4	32.4	32.8	32.1	30.3	24.5	18.4	13.6
	5%	WB	5.7	8.3	13.5	16.2	20.7	23.1	24.1	23.8	21.8	17.7	11.8	6.5
		MCDB	10.7	14.5	19.0	22.3	26.9	31.2	32.2	31.3	29.5	23.0	16.8	11.4
	10%	WB	4.0	6.4	11.1	14.7	19.6	22.4	23.5	23.0	20.9	16.1	10.1	4.8
		MCDB	8.0	11.5	17.2	21.0	25.4	30.1	31.7	30.4	27.7	21.2	15.5	8.7
Mean Daily Temperature Range	5% DB	MDBR	12.3	12.8	13.9	13.8	13.0	13.3	13.4	12.8	13.1	13.8	13.5	11.8
		MCDBR	17.9	18.5	18.9	18.2	16.1	15.8	16.1	15.3	15.2	17.4	17.7	16.4
		MCWBR	11.2	10.9	10.0	9.1	7.0	5.2	4.5	5.0	5.5	8.2	9.9	10.4
	5% WB	MCDBR	16.9	17.4	15.7	15.4	13.8	14.1	13.6	13.2	13.2	13.6	15.5	15.4
		MCWBR	10.8	10.7	9.3	8.7	7.0	5.7	5.0	5.4	5.5	7.5	9.5	10.3
Clear-Sky Solar Irradiance	taub		0.273	0.282	0.315	0.342	0.369	0.389	0.409	0.396	0.361	0.318	0.281	0.270
	taud		2.527	2.502	2.433	2.388	2.394	2.369	2.343	2.378	2.415	2.516	2.545	2.527
	Ebn at Noon		918	954	947	934	910	888	866	871	885	898	902	896
	Edh at Noon		74	87	104	116	118	121	124	117	106	86	73	69
All-Sky Solar Radiation	RadAvg		2.46	3.27	4.38	5.45	6.29	7.04	6.89	6.10	5.03	3.67	2.73	2.14
	RadStd		0.20	0.33	0.34	0.30	0.41	0.38	0.34	0.36	0.30	0.38	0.23	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page